

Question ID: 5222ffab

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	Hard

Question

While researching a topic, a student has taken the following notes:

- Neuroscientists Krishnan Padmanabhan and Zhen Chen sought to better understand the workings of the brain’s olfactory system.
- They devised a study using mathematical models.
- They found that certain fibers allow the brain to toggle from one method of processing smells to another.
- In one method, cells in the piriform cortex (where the perception of odor forms) capture olfactory information at a given moment.
- In the other, the cells track changes in olfactory information over time.

The student wants to summarize the study’s findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

- A. To arrive at these findings, which describe dual methods of processing smells in the piriform cortex, Padmanabhan and Chen devised a study using mathematical models.
- B. Padmanabhan and Chen showed that olfactory information is captured by cells in the piriform cortex, where the perception of odor forms.
- C. Using mathematical models, Padmanabhan and Chen devised a study to better understand the workings of the brain’s olfactory system.
- D. According to Padmanabhan and Chen, the brain can toggle between capturing olfactory information at a given moment and tracking changes in that information over time.

Correct Answer: D

Rationale

Choice D is the best answer. The sentence effectively summarizes the study’s findings, explaining what Padmanabhan and Chen found: that the brain can toggle between one method of processing smells (capturing information at a given moment) and another (tracking changes in information over time).

Choice A is incorrect. While the sentence mentions findings, it mainly focuses on Padmanabhan and Chen’s methodology. It doesn’t effectively summarize the study’s findings. Choice B is incorrect. The sentence notes a fact about the olfactory system—that the perception of odor forms in the piriform cortex—but doesn’t summarize the findings of Padmanabhan and Chen’s study. Choice C is incorrect. The sentence presents the goal of Padmanabhan and Chen’s study; it doesn’t summarize the study’s findings.